

- 18** A musical instrument is made using a long tube with a mouthpiece at one end. The other end is open and flared, as shown.



A musician maintains stationary sound waves with a node at the mouthpiece and an antinode at the other end. The lowest frequency of sound that the instrument can produce is 98 Hz.

Which different frequencies of sound can be produced by the instrument?

- A** 98 Hz, 147 Hz, 196 Hz, 245 Hz
- B** 98 Hz, 196 Hz, 294 Hz, 392 Hz
- C** 98 Hz, 196 Hz, 392 Hz, 588 Hz
- D** 98 Hz, 294 Hz, 490 Hz, 686 Hz

19 An optical telescope just resolves two distant yellow stars. They are subtended at an angle